Xiaotao Shen, Ph.D.
Assistant Professor
Director of Singapore Phenome Center
Lee Kong Chian School of Medicine
Nanyang Technological University, Singapore
59 Nanyang Dr., Singapore 636921
Phone: +65 83042333
Email: xiaotao.shen@ntu.edu.sg

November 26th, 2025
Editors of *Genome Biology*

Dear Editors,

We are pleased to submit our manuscript entitled "*Leveraging Large Language Models for Redundancy-Aware Pathway Analysis and Deep Biological Interpretation*" for consideration by *Genome Biology*. We specifically request that this manuscript be considered for the collection: "Application of large language models in genome analysis."

This work presents MAPA, a novel computational framework that integrates Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) to address a long-standing and pervasive problem in functional interpretation of omics data: pathway redundancy and fragmented annotation in enrichment analysis.

Pathway analysis is widely used in systems biology to uncover biological mechanisms from vast and complex omics data. However, conventional pathway analysis tools often return long, highly overlapping lists of pathways that are difficult to interpret and integrate into a cohesive biological narrative. MAPA systematically resolves this bottleneck through a two-stage process:

(1) Redundancy-aware functional module identification:

MAPA first embeds enriched pathways into a semantic space using text-transformer models. It then constructs a similarity network and applies clustering methods to group functionally similar pathways into coherent functional modules. This step transforms dozens or even hundreds of redundant pathways into unified biological functional units.

(2) Functional module annotation based on LLM and RAG:

Each functional module is then annotated using an RAG framework to reduce hallucinations. MAPA dynamically retrieves the most relevant biological descriptions of member pathways and prompts an LLM to generate accurate and concise module annotations grounded in domain knowledge. This approach not only enhances interpretability but also ensures reproducibility and transparency, as all intermediate retrievals and generations are recorded.

Our benchmarking studies show that MAPA outperforms existing tools in both functional module identification (Adjusted Rand Index = 0.95) and annotation quality (highest agreement with human expert labels across eight curated modules). To further demonstrate real-world utility, we applied MAPA to a comprehensive multi-organ monkey aging dataset, spanning transcriptomics across 78 tissues, plasma proteomics, and plasma metabolomics. MAPA uncovered robust and interpretable patterns of age-related dysregulation, such as ferroptosis vulnerability along the brain-gut axis and systemic downregulation of FOXO signaling, not captured by traditional enrichment approaches.

As LLMs continue to transform biomedical research, a growing body of work has explored their use for gene set annotation (Hu *et al.*, Nature Methods, 2024; Wang *et al.*, Nature Methods, 2025), disease modeling (Li *et al.*, Nature Medicine, 2025), and so on. Here, based on LLMs and RAG, MAPA offers a principled, task-specific, and interpretable solution to a core methodological challenge in omics: the functional redundancy of pathway analysis results.

We believe this work is ideally suited for the *Genome Biology* collection on the ‘Application of large language models in genome analysis.’ Your call for papers emphasizes the need for advanced tools that can ‘extract meaningful insights’ from complex biological data. MAPA directly addresses this goal by leveraging LLM-powered embeddings and RAG to resolve the critical bottleneck of pathway redundancy in transcriptomic, proteomic, and metabolomic datasets. By automating the functional annotation of these molecular sets, our framework transforms fragmented enrichment results into coherent biological narratives. This directly aligns with the collection’s focus on ‘LLMs for functional prediction and annotation,’ offering a powerful framework for the deep interpretation of high-dimensional omics data.

This manuscript has not been published or submitted elsewhere, and all authors have read and approved the final version. A complete set of code and documentation is available via GitHub (https://github.com/jaspershen-lab/mapa) and MAPA website (https://www.shen-lab.org/mapa-website/), ensuring full reproducibility and transparency. A web-based MAPA application is also available at: https://mapashiny.jaspershenlab.com/.

To facilitate the review process, we are happy to provide temporary OpenAI and SiliconFlow API keys for reviewers to run MAPA:

- OpenAI API key (for local version of MAPA and mapashiny):

  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

- SiliconFlow API key (for online version mapashiny):

  sk-ybmjbxlvupgvmbaqievpkntlinzeptnxmubsqsctgincxgbw

Thank you for your consideration and look forward to your response.

Yours sincerely,

Xiaotao Shen

Xiaotao Shen, Ph.D.

Director of Singapore Phenome Center,
Lee Kong Chian School of Medicine, Nanyang Technological University, Singapore.
E-mail: xiaotao.shen@ntu.edu.sg

Potential Reviewers

Include:

1. Assistant Prof. Jinzhuo Wang, Peking University, China, wangjinzhuo@pku.edu.cn (AI for biomedicine, computational biology, and multi-omics)

2. Assistant Prof. Mingxun Wang, UC Riverside, USA, mingxun.wang@ucr.edu (Computational biology and bioinformatics)

3. Associate Prof. Nima Aghaeepour, Stanford University, USA, naghaeep@stanford.edu (AI for medicine)

Xiaotao Shen, Ph.D.
Assistant Professor
Director of Singapore Phenome Center
Lee Kong Chian School of Medicine
Nanyang Technological University, Singapore
59 Nanyang Dr., Singapore 636921
Phone: +65 83042333
Email: xiaotao.shen@ntu.edu.sg

4. Dr. Mengzhou Hu, UCSD, USA, mhu@ucsd.edu (Computational biology and multi-omics)

5. Assistant Prof. Zhi Huang, University of Pennsylvania, zhi.huang@pennmedicine.upenn.edu (AI/ML innovation and its application to medicine)

6. Assistant Prof. Qiao Liu, Yale University, USA, qiao.liu@yale.edu (AI for biomedicine and computational biology)

7. Acting assistant Prof. Haoqi Zhao, Stanford University, USA, hqzhao@stanford.edu (Computational biology and multi-omics)